

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV **COURSE CODE:** DSA0202

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analysing images.. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 3 Total Marks:30 Annexure A

Scenario-Based

Code of Conduct:

I _JAASIEL J.S (192524046)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary

action. Signature of the student:

RUBRICS FOR CALCULATION

			Scenario Based Assessment			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	

Appl icati on of Conce pts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Deci sion Maki ng	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarit y of Resp onse	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganiz ed and difficult to understand	

Annexure A

Scenario-Based

1. Crack Detection on Concrete Surfaces Using Drone Images

Given:

- Images captured using a drone
- Problems: Uneven illumination, low contrast, Gaussian noise, and camera vibration
- Objective: Detect thin cracks accurately.

A. Propose a suitable image preprocessing pipeline.

Ans:

A suitable preprocessing pipeline is:

Image Acquisition → Illumination Correction → Gaussian Noise Removal → Contrast Enhancement → Image Normalization → Edge Detection

Justification:

- Illumination Correction: Removes the effect of uneven sunlight and shadows.
- Gaussian Noise Removal: Uses a Gaussian Filter to reduce image noise caused by the camera.
- Contrast Enhancement: Improves the visibility of thin cracks using techniques like Histogram Equalization or CLAHE.
- Image Normalization: Makes the pixel intensity values uniform, improving detection accuracy.

This preprocessing pipeline produces a cleaner image for crack detection.

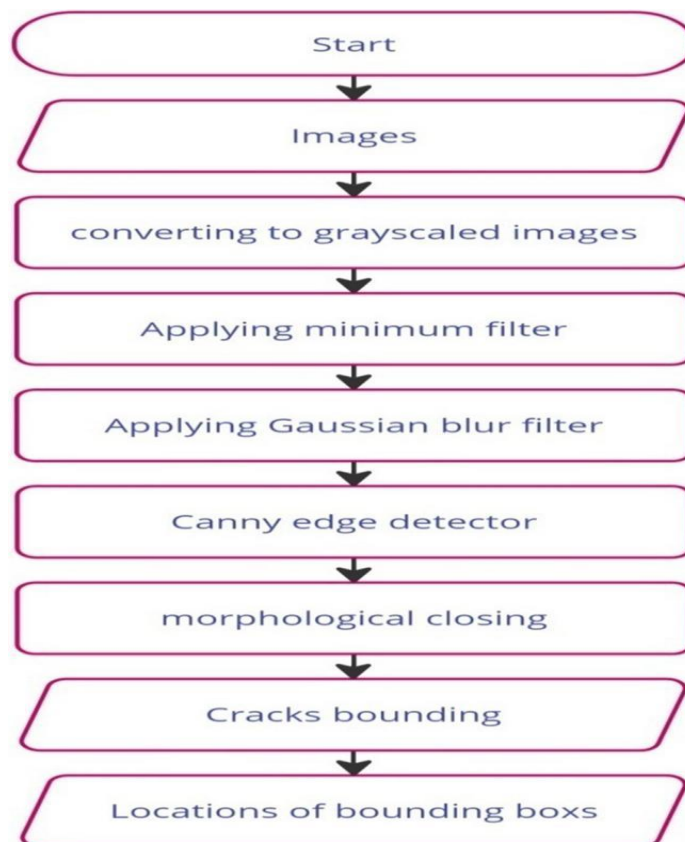


Figure 1: *Image preprocessing pipeline for drone-based crack detection.*

B. Select a suitable edge detection technique.

Ans: The Canny Edge Detector is the best choice for detecting thin surface cracks.

Comparison

Canny

Sobel

Detects thin edges accurately

Detects only strong edges

Removes noise before detection

More sensitive to noise

Produces fewer false edges

May produce unwanted edges

Ans: Canny is more suitable because it detects fine cracks accurately while reducing the effect of noise.

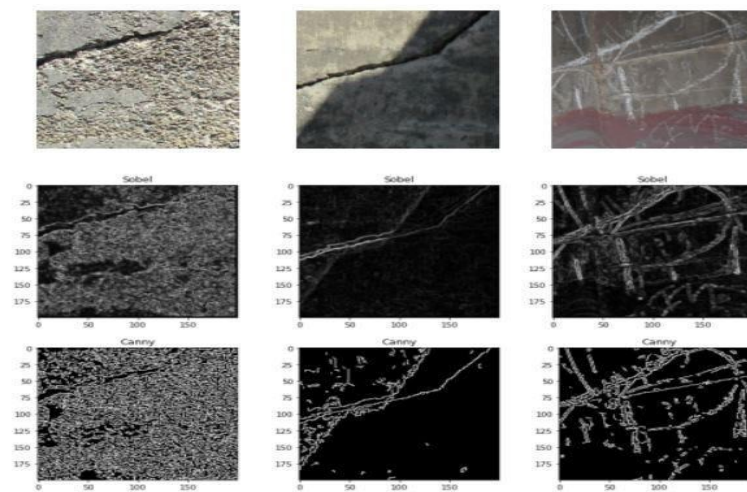


Figure 2: Comparison of Canny and Sobel edge detection for thin crack detection.

C. Suggest a post-processing technique to connect broken crack segments.

Ans: A suitable post-processing technique is Morphological Closing.

Closing connects broken crack segments and fills small gaps between crack pixels. This produces a continuous crack pattern and improves the final detection result.

2. Human Motion Tracking in a Security Camera Video

Given:

- Security camera installed in a hallway
- People move in different directions

- Hallway has plain walls, smooth floors, and shadows
- Objective: Estimate motion for tracking and activity analysis.

A. Explain how the Lucas-Kanade optical flow method can be used to estimate the motion of people between consecutive video frames.

Ans: The Lucas-Kanade Optical Flow method tracks the movement of feature points between two consecutive video frames. It assumes that the brightness of the object remains almost the same while moving. By comparing the position of feature points in consecutive frames, it calculates the direction and speed of motion. This helps in tracking people as they move through the hallway.

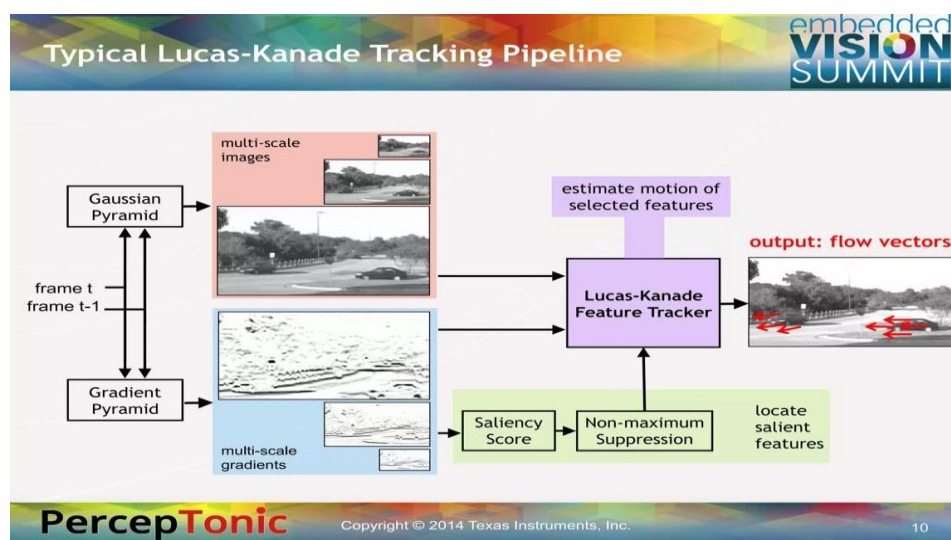


Figure 1: *Lucas-Kanade Optical Flow tracking feature points between consecutive video frames.*

B. Discuss how the aperture effect can create errors in motion estimation.

Ans: The aperture effect occurs when there are very few visible features in an image. In plain walls, smooth floors, or clothing surfaces, the optical flow algorithm cannot accurately determine the actual direction of movement. As a result, the estimated motion may be incorrect or incomplete, reducing tracking accuracy.

Optical Flow (Lucas-Kanade Method) (7/9)

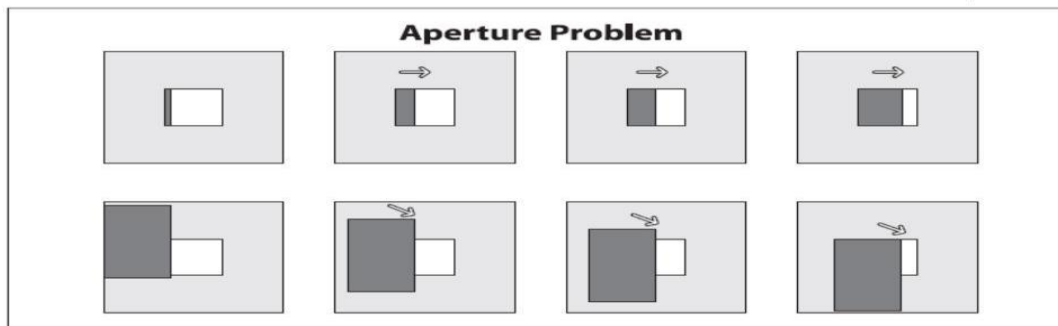


Figure 10-8. Aperture problem: through the aperture window (upper row) we see an edge moving to the right but cannot detect the downward part of the motion (lower row)

Figure 2: Aperture effect causing incorrect motion estimation in uniform regions

C. If a person suddenly changes speed or direction, describe how the optical flow vectors will be affected. Suggest a method to improve tracking accuracy.

Ans: If a person suddenly changes speed or direction, the optical flow vectors also change in both direction and magnitude. This may cause incorrect tracking for a few frames. To improve tracking accuracy, the system can use the Pyramidal Lucas-Kanade Optical Flow method, which handles larger movements more effectively and provides more reliable tracking.

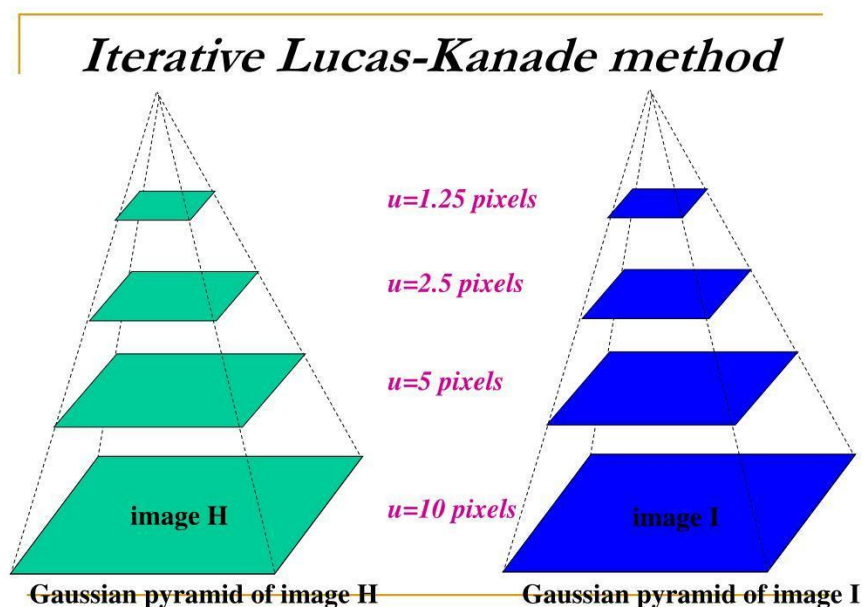


Figure 3: Pyramidal Lucas-Kanade Optical Flow for improved tracking of fast-moving objects

3. Vehicle Tracking at a Traffic Junction Using Video Frames

Given:

- CCTV camera at a busy road junction
- Vehicles move at different speeds
- Vehicles may stop, turn, or overlap
- Shadows and reflections are present
- Objective: Track vehicle motion accurately

A. Explain how optical flow can be used to estimate the direction and speed of moving vehicles in the video sequence.

Ans: Optical flow estimates the movement of vehicles by comparing consecutive video frames. It calculates the movement of feature points to find the **direction** and **speed** of each vehicle. This helps the system track vehicles as they move, stop, or turn at the road junction.

B. Identify two challenges that may affect accurate motion estimation and explain their effects.

Ans:

- **Occlusion**
When vehicles overlap, some parts of a vehicle become hidden. This makes it difficult for the optical flow algorithm to track the correct movement, leading to inaccurate motion estimation.
- **Shadows**
Moving shadows may be detected as moving objects. This creates false optical flow vectors and reduces the accuracy of vehicle tracking.

C. Suggest a suitable improvement method to make vehicle tracking more reliable when vehicles overlap or change direction suddenly

Ans: A suitable method is the **Kalman Filter**. It predicts the future position of a vehicle based on its previous movement. Even when vehicles overlap or suddenly change direction, the Kalman Filter helps maintain smooth and accurate tracking.